Mobile Application Development

Is it worth learning mobile development in 2026?

Yes

- Everyone's on their phone, and apps get the camera, GPS, Bluetooth, Face ID
- One codebase, both platforms (Flutter, React Native, KMP)
- AI edge integration is a hot new thing (on-device AI experiences).
- Mobile devs are more niche, so less competition (but also fewer job opportunities).

No

- Web apps get a lot done, no need for mobile.
- AI and no-code tools can build simple apps for you.
- App stores are a pain: reviews, fees, rules.
- Setup is a hassle (Xcode, Android Studio, emulators…)

Takeaway

> Still worth it, as long as you build what the web can't: fast apps that use the phone's hardware and on-device AI.

Popularity

- Language popularity in 2026
- Notice the percentage values of these languages: `Dart` / `Kotlin` / `Swift` / `Java` compared to `JavaScript`.
- I would say that around 10%-20% of developers (that filled the responses) are mobile devs.

Overview of Mobile Development

Mobile Development

- **Native**
  - Andriod: `Java` or `Kotlin`
  - iOS: `Objective-C` or `Swift`
- **Cross-platform**
  - React Native: `JavaScript`
  - Flutter: `Dart`
  - Kotlin Multiplatform: `Kotlin`

Native vs Cross-Platform

	Native	Cross-Platform
Time to market	Slow	Fast
Features	Full	Limited
Performance	More	Less
Cost	More	Less

Popularity

- Flutter vs React Native in 2024

> I cannot find a good source for 2026.

React Native vs Flutter

	React Native	Flutter
Language	JavaScript	Dart
UI	Native UI and iOS components	Custom widget
Dev API	Core + 3rd party libs	Core
Dev option	More versatile	More streamlined
Performance	Slower	Faster

Source

Scope of this class

- Mobile framework: `Flutter` *(will cover React Native Android and Native briefly)*
- Platform: `Android` only
  - `Android` can be developed by `Windows`, `macOS`, or `Linux`.
  - `iOS` can only be developed by `macOS`.
- Backend: `Express.js` + `PostgreSQL` + `Docker` for Developement/Deployment

Note about devices

- If you have an Android device:
  - You are good to go.
- If you have an iPhone:
  - You have to use emulator which is not always convinient or performances (memory hungry).
  - I recommend getting a cheap andriod phone.

Activities

- Preflight Project
- Quiz
- Final Project
  - Project Proposal
  - Code Reviews
  - Final Project Presentation

Schedule

Link

Preflight Project

> Develop an end-user-ready mobile full-stack app within a month.

Why

- Give students experience of the overview of the mobile technology. Before going into "real" project.
- Smaller group (1-2 people) -> better learning experience.
- More chance to iterate.

Requirement

- Frontend
  - `Flutter`
- Backend:
  - `Express.js` + `PostgreSQL` (or other backend frameworks)
  - No cloud services (Firebase, Supabase, AWS, etc.) allowed.
- Deployment:
  - Backend: `Docker` on Private VPS (CPE server)
  - Mobile: `CPE F-Droid` Store

Deliverables

- A mobile app deployed on CPE F-Droid store.
- A backend server deployed on CPE VPS.
- URL to source code repositories.
- URL to your VDO
    - File downloadable
    - Max 10 minutes long

Grade

- 20%

Due Date

- Approximately 1 month from 17 November 2026

What I am looking for

- Following the lecture examples to cover all basic requirements.
  - Already get decent grade.
- Beyond basic requirements (extra credits)
  - Good UI/UX design
  - Added functionality (e.g. authentication)
  - Added technology (e.g. libraries)

Tools (General)

- Terminal
  - (Windows) Powershell 7 + Chocolatey
  - (Mac/Linux) Zsh
- Editor
  - VSCode (and others)
- Containerization Platform
  - Docker (Desktop)

Tools (General)

- Node.js
  - If you need to use multiple versions of Node.js, consider using nvm-windows.
- Git
- Database client
  - DBeaver
- API Testing
  - Postman
  - Insomnia

Tools (Flutter)

- Flutter SDK
- Android Studio

Setup

- Windows
- macOS